ROBO DOG

Quadruped Robot — Technical Report

Arduino Uno | 10A Buck Converter | Dual HC-SR04 Ultrasonics | 8(mg90s)

GitHub: github.com/Chiraggalav0024/robo_dog

VERSION	CONTROLLER	POWER	SENSORS
1.0	Arduino Uno	10A Buck Conv.	2x HC-SR04

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1 1. PROJECT OVERVIEW

The **Robo Dog** is a four-legged (quadruped) walking robot designed and built as a DIY open-source robotics project. The robot mimics the locomotion of a dog using twelve servo motors — two per leg — controlled entirely by an **Arduino Uno** microcontroller. Two **HC-SR04 ultrasonic sensors** provide front and rear obstacle detection, and a **10A buck converter** steps down the battery voltage to a clean, regulated supply capable of driving all servos simultaneously without voltage drops.

> All source code, 3D models, and schematics are hosted publicly at:
> **https://github.com/Chiraggalav0024/robo_dog**

Key Project Highlights

Feature	Detail
Controller	Arduino Uno (ATmega328P, 16 MHz)
Locomotion	Quadruped — 4 legs × 2 DOF = 8 servo motors
Obstacle Detection	2× HC-SR04 Ultrasonic Sensors (front + rear)
Power Supply	10A Buck Converter — regulated 5V to servos & Uno
Communication	PWM signals from Arduino digital/analog pins
Programming	Arduino C/C++ — servo library + custom gait engine
Project Type	Open Source, DIY, Educational Robotics

2 2. HARDWARE COMPONENTS

2.1 Bill of Materials

Component	Qty	Specification / Notes
Arduino Uno (Rev 3)	1	ATmega328P, 14 digital I/O, 6 PWM, 16 MHz
SG90 / MG996R Servo Motor	8	3 per leg (Hip, Thigh, Knee joints)
HC-SR04 Ultrasonic Sensor	2	Range 2–400 cm, 5V, Trig + Echo pins
10A Buck Converter Module	1	Input 7–35V, Output 5V/10A, XL4016 based
LiPo Battery / 18650 Pack	1	7.4V–12V input to buck converter
Servo Extension Wires	8	Male-to-Male 3-pin servo cables
Jumper Wires	~20	Male-to-Male and Male-to-Female
Custom / 3D-Printed Frame	1	Quadruped body + 4 leg assemblies
Protoboard / PCB	1	For clean wiring connections
Power Switch	1	Main on/off for battery circuit
Capacitors (100µF, 10µF)	1	Decoupling on power rails

3 3. SYSTEM ARCHITECTURE

The diagram below shows the complete signal and power flow of the Robo Dog system. The Arduino Uno sits at the centre, receiving sensor data from both HC-SR04 modules and outputting PWM signals to all twelve servos. The 10A buck converter provides a stable 5V power rail to both the Arduino and the servo bank.

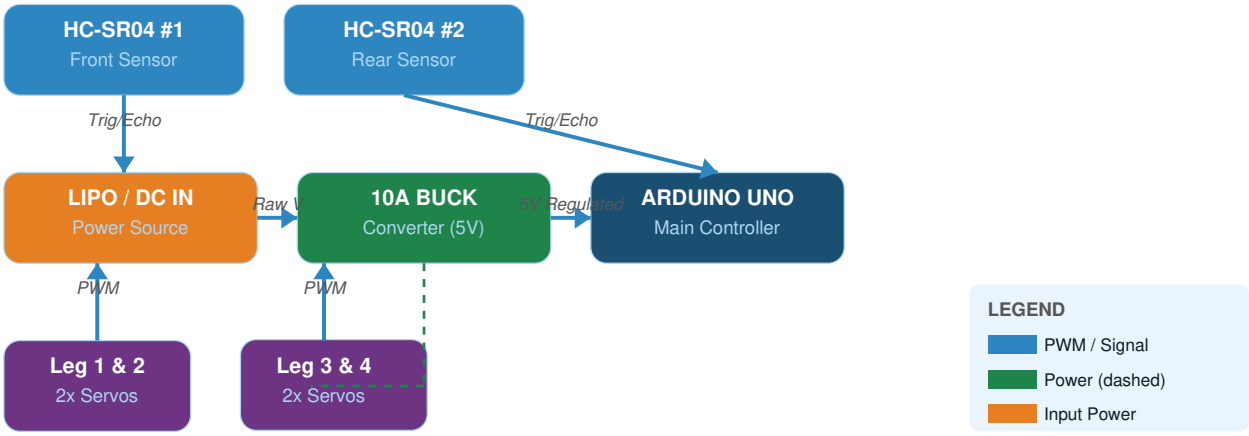

Figure 1 — System Architecture & Signal Flow

4 4. WIRING & PIN ASSIGNMENTS

4.1 Arduino Uno Pin-Out Diagram

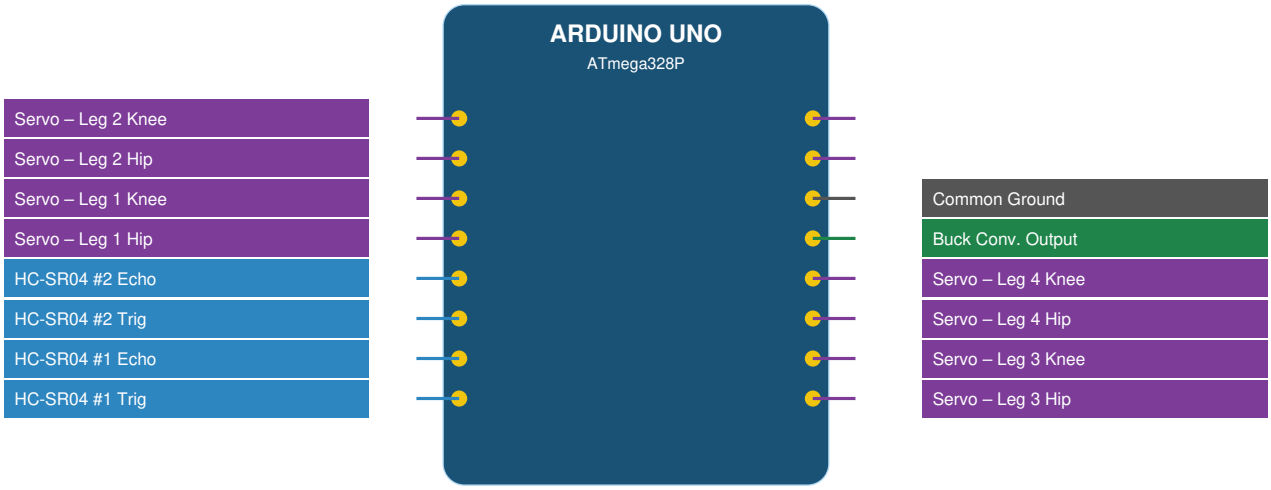

Figure 2 — Arduino Uno complete pin assignment for Robo Dog

4.2 Power Wiring

Connection	Wire	Note
Battery (+) → Buck IN+	Red, 18 AWG	Raw battery voltage (7.4–12V)
Battery (–) → Buck IN–	Black, 18 AWG	Common negative
Buck OUT+ → Arduino Vin	Red, 20 AWG	Regulated 5V to Uno
Buck OUT+ → Servo Power Rail	Red, 18 AWG	5V to all 12 servo VCC pins

Buck OUT → Arduino GND	Black, 20 AWG	Common ground
Buck OUT → Servo GND Rail	Black, 18 AWG	Servo common ground

> **Important:** Do NOT power servos from the Arduino 5V pin. The Uno can only source ~500mA which is insufficient for 12 servos. Always connect servo power directly to the buck converter output rail and share only the GND with the Arduino.

4.3 HC-SR04 Sensor Wiring

HC-SR04 Pin	Arduino Pin	Description
VCC (Sensor #1)	5V (Buck Rail)	Power supply
GND (Sensor #1)	GND	Common ground
TRIG (Sensor #1)	D2	Trigger pulse output from Arduino
ECHO (Sensor #1)	D3	Echo pulse input to Arduino
VCC (Sensor #2)	5V (Buck Rail)	Power supply
GND (Sensor #2)	GND	Common ground
TRIG (Sensor #2)	D4	Trigger pulse output from Arduino
ECHO (Sensor #2)	D5	Echo pulse input to Arduino

5 5. FIRMWARE OVERVIEW

The Arduino Uno firmware is written in C/C++ using the Arduino IDE. It handles three core responsibilities simultaneously: servo PWM generation for gait execution, ultrasonic ranging for obstacle avoidance, and state machine logic for selecting between walk, turn, and stop behaviours.

5.1 Libraries Used

Library	Purpose	Install
<Servo.h>	PWM control of all 12 servos	Built-in Arduino IDE
<NewPing.h>	HC-SR04 ultrasonic distance reading	Library Manager

5.2 Core Firmware Structure

```
// Pin Definitions
#define TRIG1 2   #define ECHO1 3   // Front HC-SR04
#define TRIG2 4   #define ECHO2 5   // Rear  HC-SR04
#define SAFE_DIST 20               // cm — obstacle threshold

Servo leg[4][3];  // leg[0..3][hip, thigh, knee]
NewPing sonar1(TRIG1, ECHO1, 200);
NewPing sonar2(TRIG2, ECHO2, 200);

void setup() {
  attachServos();   // map pins → servo objects
  standUp();        // move to neutral stance
}
```

```
void loop() {
  int front = sonar1.ping_cm();
  int rear  = sonar2.ping_cm();
  if (front > 0 && front < SAFE_DIST) { turnLeft(); }
  else if (rear > 0 && rear < SAFE_DIST) { walkForward(); }
  else { walkForward(); }
  delay(50);
}
```

5.3 Gait Engine

The gait engine implements a **trot gait** where diagonal leg pairs (Front-Left + Rear-Right, Front-Right + Rear-Left) move together. Each gait cycle consists of a **lift phase** (hip lifts leg, knee swings forward) and a **push phase** (leg pushes back, body moves forward). Servo angles are defined as arrays of position keyframes and stepped through with configurable delays.

5.4 Obstacle Avoidance Logic

Condition	Action	Notes
Front dist < 20 cm	Turn Left	Avoid forward obstacle
Front dist < 10 cm	Back Up	Very close — reverse
Rear dist < 15 cm	Walk Forward	Push away from rear obstacle
Both clear	Walk Forward	Normal locomotion
Sensor reads 0	Ignore / skip	0 = out of range or error

6 6. POWER SYSTEM — 10A BUCK CONVERTER

A key challenge in driving 12 servos from a single battery is providing adequate current without voltage sag. Each SG90/MG996R servo can draw up to 600mA–1.5A under stall load. With 12 servos, the theoretical peak current demand is **7–18A**. A 10A rated XL4016-based buck converter handles this comfortably.

Parameter	Specification
Input Voltage	7V – 35V DC
Output Voltage	1.25V – 30V (set to 5V)
Rated Current	10A continuous
Efficiency	Up to 96%
IC	XL4016 / LM2596 (common modules)
Output Ripple	< 50mV at full load
Protection	Over-temperature, short circuit

> Set the buck converter output to exactly **5.0V** using a multimeter before connecting any servos or the Arduino. Over-voltage will permanently damage SG90 servos. Adjust via the onboard trim potentiometer.

7 7. MECHANICAL DESIGN

7.1 Leg Structure (3 DOF per leg)

Each of the four legs has three joints driven by individual servo motors, providing three degrees of freedom (3-DOF). This allows complex foot trajectories needed for stable walking gaits.

Joint	Servo	Motion	Range
Hip (Lateral)	Servo A	Side-to-side swing	±30°
Knee (Sagittal)	Servo **B**	Leg extension/flex	0–90°

7.2 Frame Material & Fabrication

The body frame and leg links are designed for **3D printing in PLA**. Key structural pieces like the hip mounts use M3 brass inserts for durability. All servo horns are attached with the screws supplied with the servo motor. The body plate holds the Arduino Uno, buck converter, and battery securely with M3 standoffs.

8 8. GETTING STARTED

Step 1 — Clone the Repository	git clone https://github.com/Chiraggalav0024/robo_dog.git cd robo_dog
Step 2 — Install Libraries	Open Arduino IDE → Sketch → Include Library → Manage Libraries Search and install: NewPing by Tim Eckel
Step 3 — Set Buck Converter Output	Power the buck converter with your battery. Measure OUT+ with a multimeter. Adjust the trim pot until it reads exactly 5.00V.
Step 4 — Upload Firmware	Open robo_dog.ino in Arduino IDE Select Board: Arduino Uno | Port: your COM port Click Upload
Step 5 — Servo Calibration	Run servo_calibrate.ino to set all 12 servos to their neutral (90°) positions. Adjust the horn attachments so all legs stand vertically.
Step 6 — Power On	Connect battery → flip power switch → robot should stand up and begin walking.

9 9. TROUBLESHOOTING

Symptom	Likely Cause	Fix
Servos jitter / reset	Power sag from Arduino 5V	Power servos from buck converter directly
Robot won't stand up	Servo neutral angles wrong	Run calibration sketch, re-attach horns
Ultrasonic always reads 0	Echo pin floating	Check wiring, add 10kΩ pull-down on Echo
Arduino resets randomly	Current spike from servos	Add 100µF cap across buck output
One leg twitches	Loose servo horn screw	Tighten horn screw, re-calibrate joint

No movement at all	Wrong COM port / upload fail	Re-upload, check IDE serial port setting

10 10. FUTURE IMPROVEMENTS

Priority	Improvement	Description
High	NodeMCU / ESP32 add-on	Wireless control via ESP-NOW or WiFi (similar to hexapod)
High	IMU (MPU-6050)	Balance correction using gyroscope + accelerometer
Medium	Additional sensors	Camera module for vision, IR sensors for floor detection
Medium	Inverse Kinematics	Replace hardcoded angles with proper IK for smooth gait
Medium	Mobile app control	Bluetooth / WiFi joystick app on Android
Low	Battery monitor	Voltage divider + LCD to display remaining charge
Low	ROS integration	Connect to ROS2 for advanced navigation and mapping

11 11. REFERENCES & LINKS

- GitHub Repository: https://github.com/Chiraggalav0024/robo_dog
- Arduino Servo Library Documentation: https://www.arduino.cc/reference/en/libraries/servo/
- NewPing Library (HC-SR04): https://bitbucket.org/teckel12/arduino-new-ping/wiki/Home
- XL4016 Buck Converter Datasheet: https://www.xlsemi.com/datasheet/XL4016%20datasheet.pdf
- HC-SR04 Ultrasonic Sensor Datasheet: Available on component manufacturer websites
- Quadruped Gait Theory: 'Introduction to Robotics' — Craig, J.J.